



DEPARTMENT OF COMPUTER SCIENCE
FACULTY OF PHYSICAL SCIENCES
UNIVERSITY OF BENIN, BENIN CITY

2024/2025 BSC COMPUTER SCIENCE SECOND SEMESTER EXAMINATIONS
COURSE: COS201 - COMPUTER PROGRAMMING I

TIME: 1hr

Instruction: Shade the correct option on your OMR Answer sheet

- _____ is not a feature of Java?
(a) Object-Oriented (b) Platform Dependent (c) Secure (d) Robust (e) None of the above
- Java does not support _____.
(a) Multiple inheritance through classes (b) Object-oriented programming (c) Interface implementation (d) Exception handling (e) All of the above
- How many primitive data types are there in Java?
(a) 6 (b) 7 (c) 8 (d) 9 (e) 10
- What will be the output of the following code?

```
int x = 5;
int y = x;
x = 10;
System.out.println(y);
```

 (a) 10 (b) 5 (c) 0 (d) Compilation Error (e) Syntax Error
- Which of the following statements about reference types is true? (a) Reference types store values directly (b) Reference types are not objects (c) Reference types store memory addresses (references) (d) Reference types can't be null (e) None of the above
- Which of the following statements is true regarding procedural vs object-oriented paradigms in Java?
(a) Java does not allow procedural programming because all methods must be non-static.
(b) Procedural programming is discouraged but still syntactically valid in Java.
(c) Procedural programming requires inheritance to implement in Java.
(d) Procedural and object-oriented programming are mutually exclusive in Java (e) None of the above
- What procedural programming characteristic is demonstrated by this code?

```
public class Calculator {
    public static int sum(int a, int b) {
        return a + b;
    }
    public static void main(String[] args) {
        int result = sum(10, 20);
        System.out.println(result);
    }
}
```

 (a) Data encapsulation (b) Inheritance and polymorphism (c) Use of static functions without objects (d) Dynamic method dispatch (e) Functional programming
- _____ does not align with Java feature of being a robust language. (a) Automatic garbage collection (b) Strong memory management (c) Lack of pointer arithmetic (d) Extensive use of goto statements (e) Multithreading
- Java programs are compiled into an intermediate form called _____. (a) Machine code (b) Source code (c) Bytecode (d) Assembly code (e) Tinycode

Using the code snippet below, answer questions 10 - 11

```
import java.util.Scanner;
public class UserInputExample {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter your name: ");
        String name = scanner.nextLine();
        System.out.println("Hello, " + name);
        scanner.close();
    }
}
```

- Why is scanner.nextLine() used instead of scanner.next() in this program? (a) nextLine() ignores user input (b) nextLine() reads only the

first word (c) next line) marks the entry line including spaces (d) next line() is deprecated. (e) Don't Know why

11. Which package must be imported to use the Scanner class in Java? (a) java.io (b) java.util (c) java.lang (d) java.math (e) java.string

12. What is variable shadowing in Java? (a) When two methods have the same name but different parameters (b) When a subclass defines a field with the same name as in its superclass (c) When a constructor calls another constructor in the same class (d) When a local variable is declared inside a loop (e) None of the above

13. Which cast is allowed without an explicit cast operator? (a) int to long (b) long to int (c) Object to String (d) Double to float (e) object to int

14. What is method overloading in Java? (a) Defining multiple methods with the same name and same parameters (b) Defining methods with different names (c) Defining multiple methods with the same name but different parameters (d) Using one method in another class (e) None of the above

Consider the code snippet below and answer questions 15 to 17

```
1 public class Dog {
2     int Height;
3     private int weight;
4
5     public Dog() {
6         weight = 42;
7     }
8
9     public void cat(){
10
11     }
12
13 }
```

15. How many method(s) does the class Dog contain? (a) 1 (b) 2 (c) 3 (d) 4 (e) 5

16. What is the Access Specifier of the variable Height? (a) Public (b) Private (c) Protected (d) None of the Above (e) All of the Above

17. Which of the following is true of the class Dog?

- (i) The name of the file is Dog.java
(ii) The name of the file is unknown
(iii) The class does not have a default constructor
(iv) The variable weight is Encapsulated
(a) i Only (b) i and ii (c) iii (d) iii and iv (e) iv only

18. The code below will not run because of an error. What should be done to resolve the issue?

```
int a = 40;
short b = a;
```

- (a) Garbage Collection (b) Casting (c) Overloading (d) Polymorphism (e) Instantiation

Consider the code below and answer questions 19 to 21:

```
public class Student{
    String Name;
    private int Age;
    Student John;
    void setAge(int Age){
        this.Age=Age;
    }

    char Sex;
    int getAge (){
        Return Age;
    }

    public void createObject() {
        John = new Student();
    }
}
```

19. What type of variable is the variable, John? (a) Local Variable (b) Automatic Variable (c) Instance variable (d) Static Variable (e) None of the Above

20. Which of the following variables is an instance Variable? (a) Name (b) Age (c) Sex (d) All of the Above (e) None of the Above

21. What is the return type of the setAge method? (a) int (b) char (c) String (d) no return type (e) default

22. Identify the return type of a method that does not return any value. (a) int (b) ~~void~~ (c) double (d) return (e) None of the Above

23. What are the variables declared in a class for the use of all methods of the class called? (a) ~~Local~~ Variables (b) Instance variables (c) Reference variable (d) Class variables (e) None of the Above

24. Identify the prototype of the default constructor in the following code:

```
public class Solution {
```

(a) Solution(void) (b) ~~void Solution()~~ (c) public Solution(void) (d) public Solution() (e) None of the Above

25. Which one of the following is not an access specifier? (a) Protected (b) ~~Static~~ (c) Public (d) Private (e) None of the Above

26. Which of the following keywords allows a class to achieving multiple inheritance? (a) ~~extends~~ (b) implements (c) inherits (d) uses (e) None of the Above

27. How can you achieve polymorphism in Java? (a) By method overloading (b) By method overriding (c) ~~By both method overloading and overriding~~ (d) By extending classes (e) None of the Above

28. What happens if a method in a subclass has the same name and parameters as a method in its superclass? (a) The superclass method is executed (b) The subclass method is executed (c) A compile-time error occurs (d) The program fails to run (e) ~~The subclass method is overridden~~

29. Which of the following is valid for single inheritance? (a) ~~class SubClass extends SuperClass {}~~ (b) Subclass inherits SuperClass {} (c) class SubClass implements SuperClass {} (d) SubClass : SuperClass {} (e) Subclass extends SuperClass {}

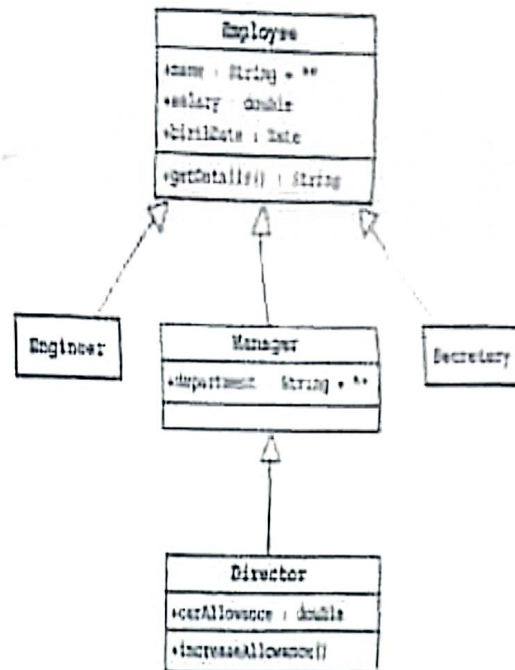
30. Which operator is used to instantiate a class? (a) extends (b) ~~new~~ (c) constructor (d) plus (e) reference

31. Find the output of the following program.

```
public class Solution {
    public static void main(String[] args) {
        short x = 10;
        x = x * 5;
        System.out.print(x);
    }
}
```

(a) ~~50~~ (b) 10 (c) 15 (d) 0 (e) Compile error

Study the diagram below and answer questions 32 and 33



32. How many variables does the Director class have? (a) 1 (b) 3 (c) 4 (d) ~~5~~ (e) 7

33. If the class defines a method, getDetails(), which of the following is an overridden method?

(a) Engineer.getDetails();
(b) Director.getDetails();
(c) Employee.getDetails();
(d) Secretary.getDetails();
(e) None of the Above

OPTION B

34. The values of an object's attributes are known as its _____. (a) state (b) orientation (c) method (d) condition (e) class
35. An instance of a class is an _____. (a) method (b) procedure (c) object (d) case (e) class
36. Java is architecturally _____. (a) neutral (b) oriented (c) specifics (d) abstract (e) object
37. All Java programming statements must end with a _____. (a) period (b) comma (c) closing parenthesis (d) semicolon (e) period
38. Arguments to methods always appear within _____. (a) parentheses (b) double quotation marks (c) single quotation marks (d) curly braces (e) square bracket
39. In a Java program, you must use _____ to separate classes, objects, and methods. (a) commas (b) semicolons (c) dots (d) forward slashes (e) apostrophe
40. Languages that let you use an easily understood vocabulary of descriptive terms, such as read, write, or add, are known as _____ languages.
- (a) procedural (b) machine (c) object-oriented (d) high-level (e) structured
41. The rules of a programming language constitute its _____. (a) logic (b) syntax (c) format (d) objects (e) byte code
42. A _____ translates high-level language statements into machine code. (a) programmer (b) syntax detector (c) compiler (d) decipherer (e) code
43. Named computer memory locations are called _____. (a) compilers (b) addresses (c) appellations (d) variables (e) bus
44. Envisioning program components as objects that are similar to concrete objects in the real world is the hallmark of _____. (a) command-line operating systems (b) procedural programming (c) object-oriented programming (d) machine languages (e) method
45. All Java applications must have a method named _____. (a) method () (b) main () (c) java () (d) Hello () (e) class ()